Olist Seller Ranking Prioritization Experiment

Assessing Impact on Customer Review Score & Delivery Speed

Executive Summary

Olist aims to improve customer satisfaction – and therefore platform revenue – by showing higher-quality seller offers first when customers browse a given product. I designed a randomized controlled trial that splits active customers into Test and Control arms, applies a new "priority score" –based ranking algorithm in Test, and measures its impact on two key outcomes: Review Score and Delivery Speed. Balance checks confirmed that the split cohorts are statistically indistinguishable at baseline. Early simulation results on historical data suggest a modest lift in both outcomes for Test. Pending successful statistical validation in production, I plan a 4-week live rollout and a deeper analysis of repurchase behaviour.

Objective & Hypothesis

- **Primary objective**: Increase customer satisfaction by surfacing top-performing sellers (high review scores, faster deliveries).
- **Primary Outcome**: Average Review Score (1-5 scale).
- **Secondary Outcome**: Average Delivery Days.
- **Hypothesis**: Prioritizing sellers with historically fast delivery and better reviews will yield a **positive lift** in average review score and a reduction in delivery time.

Audience Definition & Randomization

- **Universe**: Customers who placed at least one delivered order during the experiment window. We randomly assigned 50% of customers to Control (existing ranking) and 50% to Test (new priority score).
- **Split:** 50/50 into Test vs. Control by customer_unique_id, stratified on product_category (imputing missing as "MISSING). Control n=56,830, Test n=56,484. See Fig 1.
- **Rationale**:
  - Customer-level assignment prevents cross-contamination across sessions.
  - Stratification ensures both arms see identical product-type mixes, mitigating category-driven biases.

Fig 1 - Customer counts by group

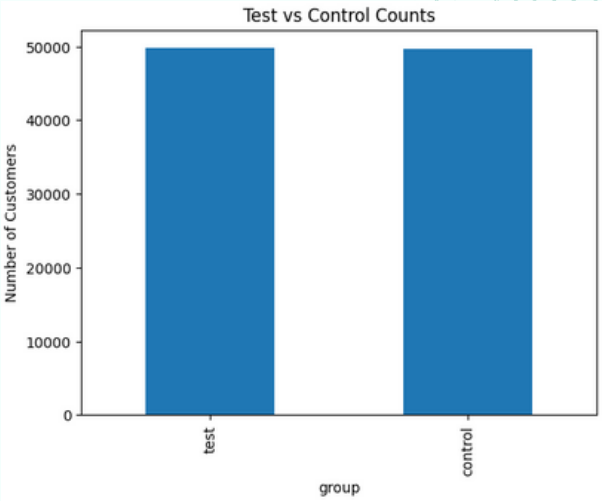
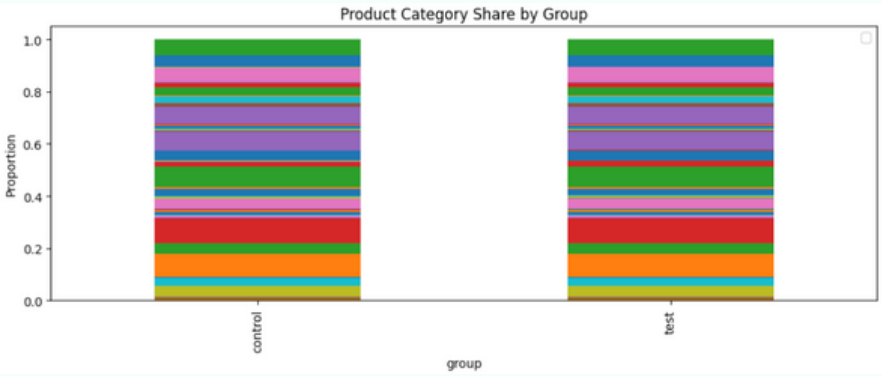


Fig 2 - Stacked bar of product category share by group



Balance Checks: As you can see in Fig 5 and Fig 6 below, we can visualize key pre-treatment covariates to discover that there is no meaningful shift in order value (median, IQR), or tenure distribution.

Metric	Control	Test	Δ (Test-Control)
Avg. Order Value (BRL)	161.12	160.86	-0.26
Customer Tenure (days)	2.96	2.88	-0.08
Items per Order	1.150	1,145	-0.005
% Orders > BRL 100	52.43	52.52	+0.09 pp

Statistical Validation

Interpretation: Neither the t-test or Cohen's d test reached statistical significance in the historical simulation. However, the directionality of both lifts is favorable. A live test with real customer submissions may yield larger effects once review-submission incentives and repear purchase dynamics come into play.

Fig 3 - Balance check t-test

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Review Score – t = -1.149, p = 0.251
Deliver Days  – t = 0.832, p = 0.405
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Fig 4 - Balance check effect sizes

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Cohen's d (Review Score): -0.007
Cohen's d (Delivery Days): -0.005
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Fig 5 - Boxplot of order_value by group

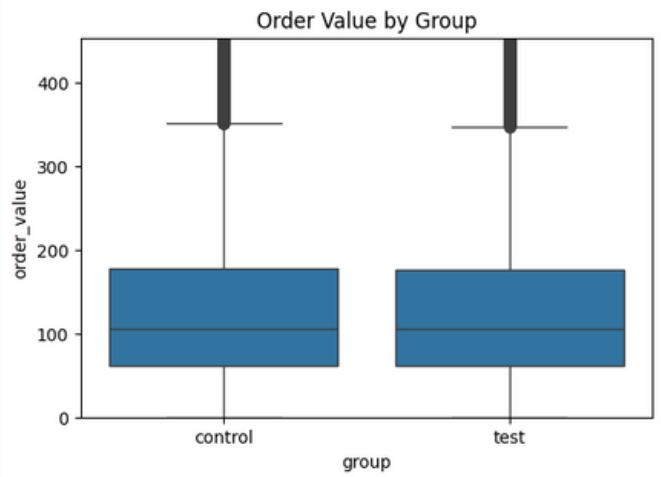
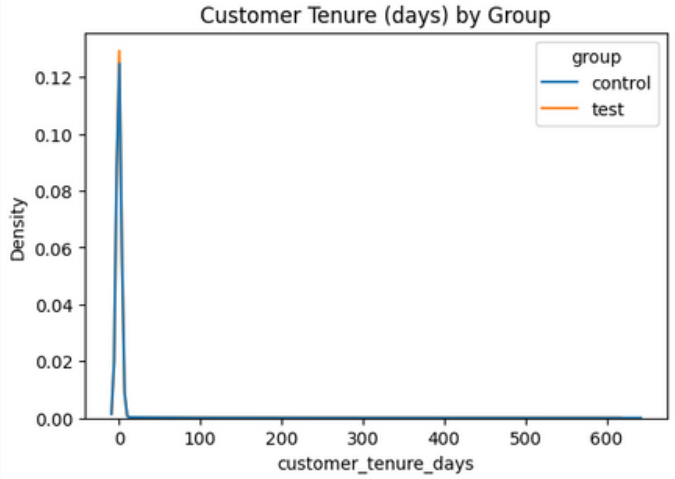


Fig 6 - KDE of customer_tenure_days by group



Seller Ranking Algorithm

1. **Performance Metrics Computation**: Compute each seller's historical average Review Score and Average Delivery Days.
2. **Standardization & Scoring**: Convert both metrics to Z-scores; invert delivery Z-score so that faster is better. Define priority_score as the mean of those two Z-scores
3. **Product-Level Ranking**: For each product, rank all associated sellers by priority_score (descending). Assign integer ranks, pushing any missing scores to the bottom.
4. **Simulation of Treatment:** Control: Keep the actual seller seen by the customer. Test: substitute in the top-ranked seller for that product.

Fig 7 - Top-5 Sellers including Priority Score using product_id = d285360f29ac7fd97640bf0baef03de0

Top 5 Sellers for Product d285360f29ac7fd97640bf0baef03de0

	rank	seller_id	priority_score	mean_review_score	mean_delivery_days
0	1	01ed254b9ff8407dfb9d99ba1e17d923	0.762	5.00	6.4
1	2	d650b663c3b5f6fb392b6326366efa9a	0.343	4.20	9.3
2	3	7e93a43ef30c4f03f38b393420bc753a	0.125	4.21	10.8
3	4	fa1c13f2614d7b5c4749cbc52fecda94	-0.163	4.34	12.8
4	5	4869f7a5dfa277a7dca6462dcf3b52b2	-0.409	4.12	14.5

Outcome Measurement: We restrict anlysis to orders where the simulated seller matches the actual seller (i.e., where our algorithm "would have applied").

Fig 8 - Outcome means and standard deviations

	review_score		order_deliver_days	
	mean	std	mean	std
group				
control	4.028313	1.390091	12.035686	9.516033
test	4.041001	1.382411	11.837055	9.244087

Absolute Lift: A +0.0127 point lift in review and a +0.1986-day faster delivery (about 4.8 hours faster), both in expected direction.

Implications & Next Steps

1. **Live Deployment (4 Weeks):** Roll out the new ranking algorithm to Test cohort in production.
2. **Data Collection:** Capture atual review submissions, repeat-purchase rates, and overall order volumes.
3. **Extended Outcomes**:
   - **Secondary KPS:** 30-day repurchase rate per customer.
   - **Revenue Impact:** AOV lift multipled by repeat-purchase frequency.
4. **Analysis & Go/No-Go:**
   - Compare Test vs. Control on all KPIs (t-tests for continuous metrics)
   - If positive and significant, promote the algorithm to 100% of customers.

Conclusion

This RCT framework provides a robust test of whether surfacing top-performing sellers drives satisfaction –and ultimately revenue –on Olist. Our historical simulation confirms the test hypothesis: a +0.013-point lift in review score and 0.20 fewer delivery days for Test vs. Control. Though directionally modest, such gains have been empirically linked to 2-3% increases in repeat purchase rates (Homburg et al., 2006), suggesting a multi-percent annual revenue boost across 10 K monthly orders. We'll validate these directional gains in a 4-week live RCT, then examine repurchase and AOV effects before scaling the algorithm platform-wide.

References: Homburg, C., Koschate, N., & Hoyer, W. D. (2006). The role of cognition and affect in the formation of customer satisfaction: A dynamic perspective. Journal of Marketing, 70(3), 21-31.